

Calculus A2 (Eng)

2025 S

Practice Midterm

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Time Limit: 3 Hours

Name: _____

Student ID: _____

This exam includes 19 pages (including this page) and 8 problems. Please check to see if there is any missing page, and then write down your name and student ID number on this page and the first page of your answer sheets. Also write down the initials of your name on the top of every page of your answer sheets, in case they are scattered.

This exam allows one SINGLE cheat sheet (size A4, double-sided is fine). No calculator or any electronic device is allowed. Plagiarism of all kinds are strictly forbidden and will be severely punished.

Please write down your answers to the problems in the **SEPARATE ANSWER SHEETS**, and follow the following rules:

- **Always explain your answer.** You should always explain your answers. At least write down one sentence to explain your thoughts.
- **Write clearly and legibly.** Make sure that your writings can be read. The graders are NOT responsible to decipher illegible writings.
- **Partial credits will be given.**
- Blank spaces are provided in the exams. Feel free to use them as scratch papers. However, your formal answer has to be written in the **SEPARATE ANSWER SHEETS**, as required by the University.
- The total score of the exam is 76. If your total score exceeds 76 (there are 80 points in total), it will be recorded as 76.

Problems	Points	Score
1	16	
2	10	
3	9	
4	9	
5	14	
6	10	
7	10	
8	10	
Total:	88	

1. (Satellite Orbit Intersection) A satellite moves along a “weird” orbit defined by the intersection of two surfaces:

$$\begin{cases} S_1 : x^2 + y^2 = 4, \\ S_2 : z = x^2 - y^2. \end{cases}$$

The intersection curve is parameterized by:

$$\gamma(\theta) = \begin{bmatrix} 2 \cos \theta \\ 2 \sin \theta \\ 4 \cos(2\theta) \end{bmatrix}.$$

- (a) (5 points) At the point corresponding to parameter $\theta = 0$, find the normal vectors for surfaces S_1 and S_2 . Calculate their cross product and explain geometrically why it has a nice relationship to $\frac{d\gamma}{d\theta}|_{\theta=0}$ of the intersection curve.
- (b) (4 points) Let us try to find the arc length of the curve for $\theta \in [0, \pi/6]$. You only need to write down the integral expression, no need to calculate the actual value.
- (c) (4 points) For the function $f(x, y, z) = xz - y$, find the derivative of the composite function $\frac{d}{d\theta}f(\gamma(\theta))$ by using chain rule.
- (d) (3 points) (Challenge) Calculate the Frenet frame $\mathbf{T}, \mathbf{N}, \mathbf{B}$ at $\theta = 0$. (Hint: Find $\gamma'(0)$ and $\gamma''(0)$ first.)

Answer:

1. Define $F(x, y) = x^2 + y^2 - 4 = 0$ and $G(x, y, z) = z - (x^2 - y^2) = 0$. At $\theta = 0$, $\gamma(0) = (2, 0, 4)$. For S_1 , $\nabla F = (4, 0, 0)$. For S_2 , $\nabla G = (-4, 0, 1)$. The cross product is $(0, -4, 0)$, which is parallel to $\gamma'(0) = (0, 2, 0)$.
2. $\gamma'(\theta) = \begin{bmatrix} -2 \sin \theta \\ 2 \cos \theta \\ -8 \sin(2\theta) \end{bmatrix}$. So $\|\gamma'(\theta)\| = \sqrt{4 + 64 \sin^2(2\theta)}$. $L = \int_0^{\pi/6} \sqrt{4 + 64 \sin^2 2\theta} \, d\theta$.
3. $[Df] = \begin{bmatrix} z & -1 & x \end{bmatrix}$. So we have

$$\frac{d}{d\theta} f(\gamma(\theta)) = \begin{bmatrix} z & -1 & x \end{bmatrix} \begin{bmatrix} -2 \sin \theta \\ 2 \cos \theta \\ -8 \sin(2\theta) \end{bmatrix} = -8 \sin \theta \cos(2\theta) - 2 \cos \theta - 16 \cos \theta \sin(2\theta).$$

4. We have $\gamma'(0) = (0, 2, 0)$ and thus we have $\mathbf{T} = (0, 1, 0)$. Then we have $\gamma''(0) = \begin{bmatrix} -2 \cos \theta \\ -2 \sin \theta \\ -16 \cos(2\theta) \end{bmatrix} \Big|_{\theta=0} = \begin{bmatrix} -2 \\ 0 \\ -16 \end{bmatrix}$, which beautifully is perpendicular to $\mathbf{T}$ already.
So $\mathbf{N} = (-\frac{1}{\sqrt{65}}, 0, -\frac{8}{\sqrt{65}})$. Finally cross product gives $\mathbf{B} = (-\frac{8}{\sqrt{65}}, 0, \frac{1}{\sqrt{65}})$.

2. (Solving Planar Intersections via Newton's Method) Consider the intersection of an ellipse and a hyperbola in the plane. We want to solve $\mathbf{F}(x, y) = \mathbf{0}$ where we define

$$\mathbf{F}(x, y) = \begin{bmatrix} \frac{x^2}{9} + y^2 - 1 \\ xy - 1 \end{bmatrix}.$$

Assume an initial guess of $(x, y) = (3, 0.5)$.

- (a) (4 points) Calculate the Jacobian matrix at the initial point and explain why $\mathbf{F}$ is differentiable there.

- (b) (2 points) Use linear approximation to estimate $\mathbf{F}(3.01, 0.49)$.

- (c) (4 points) Perform one iteration of Newton's method to find a new approximate solution.

Answer:

1. Let

$$F_1(x, y) = \frac{x^2}{9} + y^2 - 1, \quad F_2(x, y) = xy - 1.$$

The partial derivatives are:

$$\frac{\partial F_1}{\partial x} = \frac{2x}{9}, \quad \frac{\partial F_1}{\partial y} = 2y; \quad \frac{\partial F_2}{\partial x} = y, \quad \frac{\partial F_2}{\partial y} = x.$$

At point $(3, 1/2)$, the Jacobian matrix is:

$$[D\mathbf{F}(3, 1/2)] = \begin{bmatrix} \frac{2(3)}{9} & 2(1/2) \\ 1/2 & 3 \end{bmatrix} = \begin{bmatrix} 2/3 & 1 \\ 1/2 & 3 \end{bmatrix}.$$

The function $\mathbf{F}$ is differentiable at this point because all partial derivatives are polynomials, which are continuous everywhere.

2. Using linear approximation:

$$\mathbf{F}(3 + 0.01, 1/2 - 0.01) \approx \mathbf{F}(3, 1/2) + [D\mathbf{F}(3, 1/2)] \begin{bmatrix} 1/100 \\ -1/100 \end{bmatrix}.$$

First, find the value at the initial point:

$$\mathbf{F}(3, 1/2) = \begin{bmatrix} \frac{3^2}{9} + (1/2)^2 - 1 \\ 3(1/2) - 1 \end{bmatrix} = \begin{bmatrix} 1/4 \\ 1/2 \end{bmatrix}.$$

Compute the differential term:

$$\begin{bmatrix} 2/3 & 1 \\ 1/2 & 3 \end{bmatrix} \begin{bmatrix} 1/100 \\ -1/100 \end{bmatrix} = \begin{bmatrix} \frac{2}{300} - \frac{1}{100} \\ \frac{1}{200} - \frac{3}{100} \end{bmatrix} = \begin{bmatrix} \frac{2-3}{300} \\ \frac{1-6}{200} \end{bmatrix} = \begin{bmatrix} -1/300 \\ -5/200 \end{bmatrix} = \begin{bmatrix} -1/300 \\ -1/40 \end{bmatrix}.$$

Thus:

$$\mathbf{F}(3.01, 0.49) \approx \begin{bmatrix} 1/4 - 1/300 \\ 1/2 - 1/40 \end{bmatrix} = \begin{bmatrix} 37/150 \\ 19/40 \end{bmatrix}.$$

3. Newton's method formula:

$$(x, y)_{\text{new}} = \begin{bmatrix} 3 \\ 1/2 \end{bmatrix} - [D\mathbf{F}(3, 1/2)]^{-1} \mathbf{F}(3, 1/2).$$

The determinant of the Jacobian A is:

$$\det A = (2/3)(3) - (1)(1/2) = 2 - 1/2 = 3/2.$$

The inverse matrix is:

$$A^{-1} = \frac{1}{3/2} \begin{bmatrix} 3 & -1 \\ -1/2 & 2/3 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 3 & -1 \\ -1/2 & 2/3 \end{bmatrix} = \begin{bmatrix} 2 & -2/3 \\ -1/3 & 4/9 \end{bmatrix}.$$

The update step is:

$$A^{-1} \mathbf{F}(3, 1/2) = \begin{bmatrix} 2 & -2/3 \\ -1/3 & 4/9 \end{bmatrix} \begin{bmatrix} 1/4 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 1/2 - 1/3 \\ -1/12 + 2/9 \end{bmatrix} = \begin{bmatrix} 1/6 \\ 5/36 \end{bmatrix}.$$

The new iteration point is:

$$\begin{bmatrix} 3 \\ 1/2 \end{bmatrix} - \begin{bmatrix} 1/6 \\ 5/36 \end{bmatrix} = \begin{bmatrix} 17/6 \\ 13/36 \end{bmatrix}.$$

This step moves the guess toward the intersection point where $F(x, y) \approx 0$, showing the iterative improvement.

3. (Implicit Function Theorem in Enzyme Kinetics) In a biological reaction, the reaction rate R is related to substrate concentration S , inhibitor concentration I , and temperature T by:

$$S + \ln(I) + T^2 - R = 0.$$

At the point $(S, I, T, R) = (2, e, 1, 4)$, answer the following:

- (a) (3 points) Find the gradient of $F(S, I, T, R) = S + \ln(I) + T^2 - R$ at this point. Geometrically, this is a normal vector to which level set of F ?

- (b) (2 points) Treating R as an implicit function of S, I, T , find $\frac{\partial R}{\partial I}$.

- (c) (2 points) Treating I as an implicit function of S, T, R , find $\frac{\partial I}{\partial R}$.

- (d) (2 points) Discuss whether the product $\frac{\partial R}{\partial I} \frac{\partial I}{\partial R} = 1$ always holds.

Answer:

1. Calculate the partial derivatives of $F(S, I, T, R) = S + \ln(I) + T^2 - R$:

$$\frac{\partial F}{\partial S} = 1, \quad \frac{\partial F}{\partial I} = \frac{1}{I}, \quad \frac{\partial F}{\partial T} = 2T, \quad \frac{\partial F}{\partial R} = -1.$$

At the point $(2, e, 1, 4)$, we have:

$$\nabla F(2, e, 1, 4) = \left(1, \frac{1}{e}, 2, -1\right).$$

Geometrically, this gradient vector is orthogonal to the level set $F = 0$ in $\mathbb{R}^4$.

2. By the Implicit Function Theorem, if we view R as a function of (S, I, T) , the partial derivative with respect to I is given by:

$$\frac{\partial R}{\partial I} = -\frac{F_I}{F_R} = -\frac{1/I}{-1} = \frac{1}{I}.$$

At the given point where $I = e$:

$$\frac{\partial R}{\partial I} = \frac{1}{e}.$$

3. Similarly, if we view I as a function of (S, T, R) , the partial derivative with respect to R is:

$$\frac{\partial I}{\partial R} = -\frac{F_R}{F_I} = -\frac{-1}{1/I} = I.$$

At the given point:

$$\frac{\partial I}{\partial R} = e.$$

4. We always have

$$\frac{\partial R}{\partial I} \frac{\partial I}{\partial R} = \left(-\frac{F_I}{F_R}\right)\left(-\frac{F_R}{F_I}\right) = 1.$$

Using the reciprocal property of the Implicit Function Theorem, as long as the partial derivatives F_I and F_R are non-zero, the product of these inverse relationship derivatives must equal 1.

4. (Differential Properties of Vector Fields) Consider the vector field:

$$\mathbf{F}(x, y, z) = \begin{bmatrix} \sin(xz) + y^2 \\ e^y \cos z - x \\ xy + \ln(1 + z^2) \end{bmatrix}.$$

(a) (5 points) Calculate the divergence $\nabla \cdot \mathbf{F}$ and the curl $\nabla \times \mathbf{F}$.

(b) (2 points) Calculate the total derivative matrix $[D\mathbf{F}(0, 0, 1)]$.

(c) (2 points) Use the eigenvalues of this matrix to determine if a local "source" or "sink" exists near $(0, 0, 1)$.

Answer:

1. Let the components of the vector field be:

$$F_1 = \sin(xz) + y^2, \quad F_2 = e^y \cos z - x, \quad F_3 = xy + \ln(1 + z^2).$$

Divergence:

$$\nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

Calculating each term:

$$\frac{\partial F_1}{\partial x} = z \cos(xz), \quad \frac{\partial F_2}{\partial y} = e^y \cos z, \quad \frac{\partial F_3}{\partial z} = \frac{2z}{1 + z^2}.$$

Thus, the divergence is:

$$\nabla \cdot \mathbf{F} = z \cos(xz) + e^y \cos z + \frac{2z}{1 + z^2}.$$

Curl:

$$\nabla \times \mathbf{F} = \begin{bmatrix} \frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \\ \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \\ \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \end{bmatrix} = \begin{bmatrix} x - (-e^y \sin z) \\ x \cos(xz) - y \\ -1 - 2y \end{bmatrix} = \begin{bmatrix} x + e^y \sin z \\ x \cos(xz) - y \\ -1 - 2y \end{bmatrix}.$$

2. At the point $(0, 0, 1)$, we have

$$[D\mathbf{F}(0, 0, 1)] = \begin{bmatrix} 1 & 0 & 0 \\ -1 & \cos 1 & -\sin 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

3. The eigenvalues of the matrix $[D\mathbf{F}(0, 0, 1)]$ are $1, 1, \cos 1$, very easy from the block triangular structure. All are positive, so this is a source. (I write this practice exam in 2025, where linear algebra is assumed. In future years we shall only do 2×2 versions, where the eigenvalues are easy to solve, even if you did not finish your linear algebra class.)

5. (Steepest Ascent and Second Derivatives) The temperature distribution on a metal plate is given by:

$$T(x, y) = 80 - 5(x^2 + 4xy + 4y^2) + 10 \cos(x - y).$$

- (a) (4 points) Find the gradient at $(1, 1)$ and write the unit vector in the direction of steepest ascent.
- (b) (6 points) Calculate the Hessian matrix at this point. Is this a local extremum? Why?
- (c) (4 points) Does the level curve of T through $(1, 1)$ define a local function $y(x)$ in a neighborhood of the point $(1, 1)$? If not, show me why. If yes, then estimate $y(1.001)$.

Answer (Revised for point (1,1)):

1. The first-order partial derivatives are:

$$T_x = -10x - 20y - 10 \sin(x - y),$$

$$T_y = -20x - 40y + 10 \sin(x - y).$$

At the point $(1, 1)$, since $x - y = 0$, we have $\sin(0) = 0$:

$$T_x(1, 1) = -10(1) - 20(1) - 0 = -30,$$

$$T_y(1, 1) = -20(1) - 40(1) + 0 = -60.$$

The gradient vector is $\nabla T(1, 1) = (-30, -60)$. The magnitude is $\|\nabla T\| = \sqrt{(-30)^2 + (-60)^2} = \sqrt{900 + 3600} = \sqrt{4500} = 30\sqrt{5}$. The unit vector in the direction of steepest ascent is:

$$\mathbf{u} = \frac{(-30, -60)}{30\sqrt{5}} = \left(-\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}} \right).$$

2. The second-order partial derivatives at $(1, 1)$ where $\cos(0) = 1$:

$$T_{xx} = -10 - 10 \cos(0) = -20,$$

$$T_{yy} = -40 - 10 \cos(0) = -50,$$

$$T_{xy} = -20 + 10 \cos(0) = -10.$$

The Hessian matrix at $(1, 1)$ is:

$$H_T(1, 1) = \begin{bmatrix} -20 & -10 \\ -10 & -50 \end{bmatrix}.$$

Since $\nabla T(1, 1) \neq (0, 0)$, the point is not a critical point and thus not a local extremum.

3. According to the Implicit Function Theorem, the equation $T(x, y) = 45$ defines a local function $y(x)$ near $(1, 1)$ if the partial derivative with respect to the dependent variable is non-zero:

$$\left. \frac{\partial T}{\partial y} \right|_{(1,1)} = -60 \neq 0.$$

Since $T_y(1, 1) \neq 0$, such a local function $y(x)$ exists. And by implicit function theorem we have

$$\frac{dy}{dx} = -\frac{T_x}{T_y} = -\frac{-30}{-60} = -\frac{1}{2}.$$

Hence we have

$$y(1.001) \approx y(1) + \left. \frac{dy}{dx} \right|_{x=1} 0.001 = 0.9995.$$

6. (Analysis Across a Boundary Curve) Consider the function $f(x, y)$ defined by different rules on either side of the parabola $y = x^2$:

$$f(x, y) = \begin{cases} x^2y, & y \geq x^2 \\ x^4, & y < x^2 \end{cases}$$

- (a) (3 points) Prove that $f(x, y)$ is continuous at every point (x_0, y_0) on the boundary curve $y = x^2$.
- (b) (3 points) Calculate the first-order partial derivatives f_x and f_y at the point $(1, 1)$.
- (c) (4 points) Prove that our function is differentiable at the point $(0, 0)$.

Answer:

1. At a point (x_0, y_0) where $y_0 = x_0^2$, we have

$$\lim_{(x,y) \rightarrow (x_0,y_0)} x^2 y = x_0^2 y_0 = x_0^4,$$

$$\lim_{(x,y) \rightarrow (x_0,y_0)} x^4 = x_0^4.$$

Since this is true for both pieces, we have

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y) = x_0^4 = f(x_0, y_0).$$

Hence continuous.

2. At $(1, 1)$, the point lies on the boundary $y = x^2$.

For f_y : We move along the vertical line $x = 1$. When y increases from 1, the function is $f(1, y) = 1^2 y = y$, and the derivative is 1. When y decreases from 1, the function is $f(1, y) = 1$, and the derivative is zero. Hence f_y fails to exist.

For f_x : We move along $y = 1$. When x increases from 1, the function is $f(x, 1) = x^4$, and the derivative is 4. When x decreases from 1, the function is $f(x, 1) = x^2$, and the derivative is 2. Hence f_x fails to exist.

3. At $(0, 0)$, first let us compute the partial derivatives.

For f_y : We move along the vertical line $x = 0$. When y increases from 0, the function is $f(0, y) = 0^2 y = 0$, and the derivative is zero. When y decreases from 0, the function is $f(0, y) = 0$, and the derivative is zero. Hence $f_y = 0$.

For f_x : We move along $y = 0$. When x increases from 0, the function is $f(x, 0) = x^4$, and the derivative is $4x^3 = 0$. When x decreases from 0, the function is $f(x, 0) = 0$, and the derivative is 0. Hence $f_x = 0$.

These mean that IF we are differentiable, then the derivative should be $\begin{bmatrix} 0 & 0 \end{bmatrix}$. Now we compute

$$\begin{aligned} |f(d\mathbf{x}, d\mathbf{y}) - f(0, 0) - \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} d\mathbf{x} \\ d\mathbf{y} \end{bmatrix}| &= |f(d\mathbf{x}, d\mathbf{y})| \\ &\leq \max(|dx^2 dy|, |dx^4|) \\ &\leq \max(\|d\mathbf{x}\|^3, \|d\mathbf{x}\|^4). \end{aligned}$$

So we have $f(d\mathbf{x}) = f(0, 0) + \begin{bmatrix} 0 & 0 \end{bmatrix} d\mathbf{x} + o(\|d\mathbf{x}\|)$. Hence we are indeed differentiable, as the Taylor expansion of degree one is true.

7. Consider the function $f(x, y)$ defined on either side of the y -axis ($x = 0$):

$$f(x, y) = \begin{cases} e^x \cos y, & x \geq 0 \\ (1 + x + \frac{1}{2}x^2) \cos y, & x < 0 \end{cases}$$

(a) (3 points) Prove that $f(x, y)$ is continuously differentiable.

(b) (4 points) Calculate the mixed partial derivatives $D_2D_1f(0, y)$.

(c) (3 points) Does Clairaut's Theorem apply on the line $x = 0$? Show why or why not. Then compute the value of D_1D_2f at $(0, y)$.

Answer:

1. First we need to show that f is continuous. On the boundary $(0, y_0)$, direct computation shows that both pieces have the same limit $\cos y_0$. So f is continuous.

It is enough to show that D_1f, D_2f are differentiable. We know $D_1f(x, y) = e^x \cos y$ for $x > 0$ and $D_1f(x, y) = (1 + x) \cos y$ for $x < 0$. On the boundary where $x = 0$, x moving above 0 gives derivative $\cos y$, and x moving below 0 gives derivative $\cos y$. So on the boundary we have $D_1f(0, y) = \cos y = e^x \cos y = (1 + x) \cos y$. Now, on both pieces and on the boundary, direct computation shows that the limits to $(0, y_0)$ will be the same $\cos y_0$, hence D_1f is continuous.

Similarly, $D_2f(x, y) = -e^x \sin y$ for $x > 0$ and $D_2f(x, y) = -(1 + x + \frac{1}{2}x^2) \sin y$ for $x < 0$. On the boundary where $x = 0$, if we move in the y direction, we stayed on the line $x = 0$ and hence we always have $f(x, y) = e^x \cos y$, so we have $D_2f(x, y) = -e^x \sin y$. In particular, on the boundary we have $D_2f(0, y) = -\sin y = -e^x \sin y = -(1 + x + \frac{1}{2}x^2) \sin y$. Now, on both pieces and on the boundary, direct computation shows that the limits to $(0, y_0)$ will be the same $-\sin y_0$, hence D_2f is continuous.

So D_1f, D_2f are continuous on the entire domain (an open set). By total differentiability theorem, this means that $f(x, y)$ is continuously differentiable on the entire domain.

2. On the line $x = 0$, $D_1f(0, y) = \cos y$. Then $D_2D_1f(0, y) = -\sin y$.
3. We need D_2D_1f continuous on the line $x = 0$. Away from the line $x = 0$, we have

$$D_2D_1f(x, y) = -e^x \sin y.$$

As $(x, y) \rightarrow (0, y_0)$, limit on one piece (the line) $x = 0$ is $-\sin y_0$ and limit on the other piece (not on the line) is $-(e^0) \sin y_0 = -\sin y_0$. This is exactly the value of $D_2D_1f(0, y_0)$. So D_2D_1f is continuous everywhere. Hence Clairaut's Theorem applies, and $D_1D_2f(0, y) = D_2D_1f(0, y) = -\sin y$.

8. (Total Derivative of a Matrix-Valued Function) Consider the function $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^{2 \times 2}$ defined by:

$$\mathbf{F}(x, y, z) = \begin{bmatrix} x^2 + y & \sin(xz) \\ e^{y+z} & xyz \end{bmatrix}$$

- (a) (4 points) Calculate the partial derivative matrices $D_1\mathbf{F}$, $D_2\mathbf{F}$, and $D_3\mathbf{F}$ at a general point (x, y, z) . Where is $\mathbf{F}$ differentiable?

- (b) (3 points) At the point $P = (1, 0, 0)$, find the directional derivative $D_{\mathbf{v}}\mathbf{F}$ with $\mathbf{v} = (1, -1, 1)$.

- (c) (3 points) Let $\mathbf{G}(P) = (\mathbf{F}(P))^2$, where the square means the square of a matrix. At the point $P = (1, 0, 0)$, find the directional derivative $D_{\mathbf{v}}\mathbf{G}$ with $\mathbf{v} = (1, -1, 1)$.

Answer:

1. The partial derivative of a matrix-valued function is calculated by taking the partial derivative of each entry:

$$\frac{\partial \mathbf{F}}{\partial x} = \begin{bmatrix} 2x & z \cos(xz) \\ 0 & yz \end{bmatrix}, \quad \frac{\partial \mathbf{F}}{\partial y} = \begin{bmatrix} 1 & 0 \\ e^{y+z} & xz \end{bmatrix}, \quad \frac{\partial \mathbf{F}}{\partial z} = \begin{bmatrix} 0 & x \cos(xz) \\ e^{y+z} & xy \end{bmatrix}$$

Everything is continuous, so by total differentiability theorem $\mathbf{F}$ is differentiable everywhere.

2. Evaluating at $P = (1, 0, 0)$, where $x = 1, y = 0, z = 0$:

$$\frac{\partial F}{\partial x}(1, 0, 0) = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}, \quad \frac{\partial F}{\partial y}(1, 0, 0) = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \quad \frac{\partial F}{\partial z}(1, 0, 0) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

So we have

$$D_{(1,-1,1)}\mathbf{F}(P) = 1 \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} - 1 \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

3. Let $h(A) = A^2$ be the map on matrices. Then we know from class (or by easy calculation of $(A + dX)^2$) that $[Dh(A)]$ is the map sending X to $AX + XA$. Now by chain rule,

$$D_v \mathbf{G} = [D\mathbf{G}](v) = [Dh(\mathbf{F}(P))]([D\mathbf{F}(P)]v) = [Dh(\mathbf{F}(P))](D_v \mathbf{F}(P)).$$

Now $\mathbf{F}(P) = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$. So we have

$$D_v \mathbf{G} = [Dh(\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix})](\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}) = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}.$$